



Multiple Choice:

1. Which of the following gases is primarily responsible for the greenhouse effect?
 - A. Carbon dioxide (CO₂)
 - B. Nitrogen (N₂)
 - C. Oxygen (O₂)
 - D. Methane (CH₄)
2. Which of the following is a renewable energy source?
 - A. Coal
 - B. Natural gas
 - C. Solar power
 - D. Nuclear power
3. The process of converting waste materials into reusable materials is called:
 - A. Recycling
 - B. Composting
 - C. Incineration
 - D. Landfilling
4. Which of the following is a non-point source of water pollution?
 - A. Industrial discharge pipe
 - B. Sewage treatment plant
 - C. Oil spill from a ship
 - D. Runoff from agricultural fields
5. What is the primary cause of ozone depletion in the Earth's stratosphere?
 - A. Carbon monoxide (CO)
 - B. Chlorofluorocarbons (CFCs)
 - C. Nitrous oxide (N₂O)
 - D. Sulfur dioxide (SO₂)
6. Which of the following is an example of a point source of air pollution?
 - A. Vehicle exhaust emissions
 - B. Dust storm
 - C. Wildfire
 - D. Volcanic eruption
7. Which of the following is a greenhouse gas that is naturally produced by wetlands and livestock?
 - A. Carbon monoxide (CO)
 - B. Nitrogen oxide (NO_x)
 - C. Methane (CH₄)
 - D. Sulfur dioxide (SO₂)
8. Which of the following is a greenhouse gas that is primarily released through the burning of fossil fuels?
 - A. Methane (CH₄)
 - B. Carbon dioxide (CO₂)
 - C. Nitrous oxide (N₂O)
 - D. Ozone (O₃)
9. What is the term used to describe the gradual increase in the Earth's average temperature due to human activities?
 - A. Global warming
 - B. Ozone depletion
 - C. Acid rain
 - D. Desertification
10. Which of the following is an example of a non-renewable resource?
 - A. Solar energy
 - B. Wind energy
 - C. Natural gas
 - D. Geothermal energy
11. What is the process by which fertile land becomes a desert, typically as a result of deforestation, overgrazing, or climate change?
 - A. Desertification
 - B. Eutrophication
 - C. Acidification
 - D. Biomagnification
12. Which of the following is a primary source of indoor air pollution?
 - A. Volcanic eruptions
 - B. Forest fires
 - C. Vehicle emissions
 - D. Tobacco smoke
13. What is the primary source of mercury pollution in aquatic ecosystems?
 - A. Industrial emissions
 - B. Agricultural runoff
 - C. Burning of fossil fuels
 - D. Artisanal gold mining
14. Which of the following is a renewable resource derived from organic matter, such as plant material and animal waste?
 - A. Geothermal energy
 - B. Solar energy
 - C. Biomass
 - D. Hydroelectric power
15. What is the primary cause of depletion in the ozone layer?
 - A. Carbon dioxide emissions
 - B. Deforestation
 - C. Volcanic eruptions
 - D. Chlorofluorocarbons (CFCs)
16. Which of the following is an example of a non-point source of water pollution?
 - A. Industrial wastewater discharge
 - B. Sewage treatment plant effluent
 - C. Oil spill from a tanker
 - D. Agricultural runoff
17. Which of the following is a renewable source of energy derived from the heat stored within the Earth's crust?
 - A. Solar power
 - B. Wind power
 - C. Geothermal energy
 - D. Hydroelectric power
18. Which of the following is a method used to reduce or prevent soil erosion?
 - A. Afforestation
 - B. Overgrazing
 - C. Deforestation
 - D. Monoculture farming
19. Which of the following is a primary greenhouse gas responsible for trapping heat in the Earth's atmosphere?
 - A. Oxygen (O₂)
 - B. Carbon dioxide (CO₂)



- C. Nitrogen (N₂)
D. Argon (Ar)
20. Which of the following is an example of a point source of air pollution?
- A. Vehicle emissions
B. Dust storms
C. Forest fires
D. Volcanic eruptions
21. What is the process by which a body of water becomes enriched with nutrients, leading to excessive algae growth and oxygen depletion?
- A. Desertification
B. Eutrophication
C. Acidification
D. Biomagnification
22. Which of the following best describes the concept of "ecological footprint"?
- A. The measure of biodiversity within a specific ecosystem
B. The total amount of land and water area required to sustain a population's consumption and waste production
C. The rate at which a population exceeds the carrying capacity of its environment
D. The measure of energy efficiency in a given ecosystem
23. Which of the following best describes the "tragedy of the commons"?
- A. The depletion of common-pool resources due to the self-interest of individuals
B. The destruction of biodiversity in common ecosystems
C. The overuse of renewable resources leading to their extinction
D. The degradation of public lands due to industrial activities
24. What is the term used to describe the long-term changes in temperature, precipitation patterns, and wind conditions in a particular region?
- A. Climate change
B. Weather variability
C. Atmospheric oscillation
D. Ecological succession
25. Which of the following is a critical factor contributing to the process of desertification?
- A. Excessive rainfall
B. Decreased wind erosion
C. Increased vegetation cover
D. Unsustainable agricultural practices
26. What is the main driver of deforestation in tropical rainforests?
- A. Climate change
B. Mining activities
C. Commercial agriculture
D. Urbanization
27. What is the primary cause of ocean acidification?
- A. Pollution from industrial activities
B. Overfishing and depletion of fish stocks
C. Invasive species in marine ecosystems
D. Increased carbon dioxide absorption by seawater
28. Which of the following pollutants is primarily responsible for causing smog in urban areas?
- A. Sulfur dioxide (SO₂)
B. Nitrogen oxides (NO_x)
C. Carbon monoxide (CO)
D. Particulate matter (PM)
29. What is the term used to describe the process of converting waste materials into organic matter that can be used as fertilizer?
- A. Incineration
B. Composting
C. Recycling
D. Landfilling
30. What is the term used to describe the loss of a species from a particular habitat or from the entire planet?
- A. Extinction
B. Endemism
C. Fragmentation
D. Biodiversity
31. Which of the following is a method used to reduce soil erosion and improve soil fertility by planting different crops in alternating rows or bands?
- A. Crop rotation
B. Monoculture farming
C. Overgrazing
D. Deforestation
32. What is the term used to describe the process of movement of individuals into a population?
- A. Emigration
B. Immigration
C. Natality
D. Mortality
33. What is the term used to describe the variety of different species in an ecosystem?
- A. Biodiversity
B. Endemism
C. Keystone species
D. Ecotone
34. What is the term used to describe the process by which water vapor is released from the leaves of plants into the atmosphere?
- A. Transpiration
B. Condensation
C. Evaporation
D. Precipitation
35. Which of the following is a renewable natural resource?
- A. Coal
B. Natural gas
C. Timber
D. Petroleum



36. What is the term used to describe the total amount of a resource that exists in the Earth's crust?
- A. Reserves
 - B. Stock
 - C. Renewable resource
 - D. Nonrenewable resource
37. Which of the following best describes a wetland ecosystem?
- A. A terrestrial ecosystem characterized by high temperatures and low rainfall
 - B. An aquatic ecosystem characterized by flowing water and diverse fish species
 - C. A land area that is saturated or flooded for a significant part of the year
 - D. A forest ecosystem dominated by coniferous trees and acidic soil
38. What is the term used to describe the excessive growth of algae in a body of water, often caused by nutrient pollution?
- A. Erosion
 - B. Acidification
 - C. Eutrophication
 - D. Desertification
39. What is the term used to describe the accumulation of toxic substances in the tissues of organisms as they move up the food chain?
- A. Biomagnification
 - B. Eutrophication
 - C. Acidification
 - D. Desertification
40. Which of the following is a consequence of deforestation?
- A. Reduced soil erosion
 - B. Increased carbon sequestration
 - C. Loss of biodiversity
 - D. Enhanced water retention
41. Which of the following is an example of a direct environmental impact?
- A. Carbon emissions from power plants
 - B. Loss of biodiversity due to habitat destruction
 - C. Acid rain caused by air pollution
 - D. Soil erosion from deforestation
42. What is the primary objective of an environmental impact assessment (EIA)?
- A. To prevent all environmental impacts
 - B. To minimize and mitigate environmental impacts
 - C. To ignore environmental impacts for economic growth
 - D. To maximize environmental impacts for sustainability
43. Which of the following is a qualitative method used in environmental impact assessment?
- A. Cost-benefit analysis
 - B. Life cycle assessment
 - C. Geographic Information System (GIS)
 - D. Expert judgment
44. Which of the following is a social impact that may be considered in environmental impact assessment?
- A. Noise pollution
 - B. Soil erosion
 - C. Water quality
 - D. Displacement of local communities
45. What is the term used to describe the process of identifying and assessing potential environmental impacts before a project is implemented?
- A. Environmental monitoring
 - B. Post-project evaluation
 - C. Environmental risk assessment
 - D. Pre-project assessment
46. Which of the following is a key principle of sustainable development?
- A. Economic growth at all costs
 - B. Exploitation of natural resources
 - C. Preservation of biodiversity
 - D. Short-term profits
47. What is the term used to describe the practice of meeting the needs of the present generation without compromising the ability of future generations to meet their own needs?
- A. Intergenerational equity
 - B. Sustainable consumption
 - C. Environmental stewardship
 - D. Ecological footprint
48. What is the concept of "triple bottom line" in sustainable development?
- A. Balancing economic, social, and environmental goals
 - B. Prioritizing economic growth over social and environmental concerns
 - C. Focusing solely on economic profitability
 - D. Promoting environmental conservation at the expense of economic development
49. What is the term used to describe the meeting of present needs without compromising the ability of future generations to meet their needs?
- A. Sustainable development
 - B. Economic growth
 - C. Environmental conservation
 - D. Resource depletion
50. What is the goal of the circular economy concept?
- A. To minimize waste and maximize resource efficiency
 - B. To promote linear production and consumption patterns
 - C. To prioritize economic growth over environmental concerns
 - D. To deplete natural resources for short-term gains